Anas Khan

AI & Full Stack Engineer · Team Lead, Python & AI

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SUMMARY

Python and AI engineering lead with 6+ years at Tassaract, promoted from Full Stack Developer to Senior Full Stack Developer to Team Lead of the Python & AI team. Builds production backends with FastAPI, Flask and Django on PostgreSQL; automation and trading bots on exchange, DEX and broker APIs; and applied-AI pipelines spanning OpenAI fine-tuning, LLM document generation, transformer-based NER and OCR/computer vision. Owns delivery end to end, from architecture and data modelling to Docker deployment, and leads the developers who build it.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS

Backend & APIs: FastAPI, Flask, Django, REST API design, Pydantic, SQLAlchemy, Alembic, JWT / OAuth2 auth and role-based access, Socket.IO / WebSockets, GraphQL (client)

AI / ML: OpenAI API (fine-tuning, Chat Completions), prompt engineering, Hugging Face Transformers (BERT NER), Flair, OpenCV, Tesseract OCR, pandas, NumPy, machine learning

Data & Infrastructure: PostgreSQL, MySQL, SQLite, AWS S3, Docker, Linux (Ubuntu, systemd, Bash)

Automation: Selenium, web scraping (cloudscraper, BeautifulSoup), IMAP and PDF parsing (pdfminer), schedulers and background workers, bot development

Trading & Blockchain: Arbitrage bots, ccxt, Solana DeFi APIs (Jupiter, Birdeye), IG Markets API (Lightstreamer), EVM wallets (BIP-39, eth-account)

Integrations: Stripe, Firebase Cloud Messaging, Google Sheets and Drive APIs, eBay Browse API, Telegram Bot API, SMTP, PayPal

Frontend: React.js, JavaScript, HTML, CSS, Bootstrap, Materialize

Leadership: Team leadership, AI team management, project planning, code review, mentoring

PROFESSIONAL EXPERIENCE

Team Lead, Python & AI Team · Tassaract — **2024 - Present**

- Lead a team of Python and AI developers: project planning, task breakdowns, code reviews and on-time delivery.
- Design and build full stack web applications in Python (Flask, Django), from database design through to the final UI.
- Guide machine learning work from data analysis and model building into applications the business uses day to day.
- Add AI features with the OpenAI API, including chatbot-style features that handle user queries automatically.
- Oversee automation and bot projects, from internal workflow bots to arbitrage bots that act on price differences across platforms, kept running through APIs, schedulers and background jobs.
- Improved the team's web scraping and automation so data extraction is faster, more reliable and needs less manual cleanup.
- Mentor junior developers through code review and hands-on guidance, helping them grow into more senior roles.

Senior Full Stack Developer · Tassaract — **2021 - 2024**

- Built and maintained full stack web applications with Python, Flask and Django, across backend logic and front end.
- Wrote scraping and automation scripts that pull data from multiple sources, cutting manual data entry.
- Built automation bots for repetitive internal tasks, including early arbitrage bots that tracked prices across platforms.
- Delivered blockchain features: wallet integration, on-chain transaction handling and smart-contract interactions inside web applications.
- Built an OpenAI-powered chatbot for common customer and internal queries, with conversation flows that hand off to a human when needed.
- Built responsive front ends with React.js, HTML, CSS and JavaScript; delivered ML projects from data analysis to deployed models.
- Built internal business applications for other departments, working directly with their users to fit real workflows.

Full Stack Developer · Tassaract 2019 - 2021

- Built and maintained web applications using Python with Flask and Django, and front-end pages with React.js, HTML, CSS and JavaScript.
- Wrote first automation bots and scraping scripts to collect and process data; supported early blockchain experiments around wallet connections and transaction testing.
- Learned full stack development on the job from senior engineers, which led to two promotions.

SELECTED PROJECTS

NicheTRX: Freight Marketplace Platform · Independent / Freelance

FastAPI, PostgreSQL, SQLAlchemy, Socket.IO, Stripe, AWS S3, Firebase, Docker, React

- Designed and built a GCC truck-booking marketplace end to end: 96 REST endpoints across 7 routers and 18 data models for clients, vendors (fleet and individual owners), drivers and admins, with KYC uploads and approval.
- Modelled the full job lifecycle: admin-moderated posting, vendor bidding, admin-set pricing within route-rate bands, two-sided client/driver confirmation, an enforced 5-step delivery tracking flow with photo proof, and Stripe Checkout confirmed by a signature-verified webhook.
- Added live GPS tracking over Socket.IO rooms, Firebase push notifications and private S3 storage with presigned URLs; containerised with Docker (database readiness check, Alembic migrations, Uvicorn).

Support Assistant Fine-Tuning Pipeline

OpenAI API, MySQL, Flask

- Built an ETL from a client's osTicket MySQL helpdesk into OpenAI chat-format training data: 5,135 messages from 789 threads, vehicle/ECU context from custom form fields, thread-level split (380 train / 50 validation).
- Automated GPT-3.5-turbo fine-tuning (file upload, job creation, status polling) on a daily schedule, with a feedback endpoint that adds star-rated replies to the next training run.

Sports Card Grading-Arbitrage Finder

eBay Browse API, Hugging Face Transformers, Flair, pandas, SQLite, Google Drive API

- Built a scanner on the eBay Browse API (OAuth2, rotation across 6 keysets) that stored 26,006 listings; a BERT-large NER model with a Flair fallback recovered player names for 884 of 1,372 listings that lacked them.
- Matched identical cards and ranked raw vs. grade-9 / grade-10 (PSA, BGS, SGC) price spreads, publishing the reports to Google Sheets with email alerts.

Solana Arbitrage, Sniper & Auto-Sell Engine

Python, Jupiter API, Birdeye WebSocket, SQLite, Flask, Telegram

- Built three concurrent engines (triangular arbitrage on Jupiter quotes, a new-token sniper on a Birdeye WebSocket feed, and a take-profit auto-seller) with graceful shutdown and Telegram alerts.
- Added risk gates (10-second quote expiry, re-quote validation with price-impact caps, a 7-rule token safety screen), a SQLite ledger with 69 audit event types and a Flask dashboard; paper-trades on live quotes.

Broker Email to TradingView Automation

IMAP, pdfminer, Selenium, cryptography, systemd

- Delivered a client automation in which an IMAP watcher parses broker trade-confirmation emails and PDFs and keeps TradingView watchlists and chart price markers in sync with trades, including partial fills.
- Built resilient Selenium tooling (multi-strategy selectors, double-verified actions), encrypted session reuse with 2FA- and captcha-capable login, health checks and a systemd service for Ubuntu.

Geo-Availability QA Bot with OCR and Computer Vision

Selenium, Tesseract OCR, OpenCV, AWS S3, Google Sheets

- Built a bot that launches each catalogue game from 20 countries via geo-targeted remote browsers and classifies the outcome with DOM checks, Tesseract OCR and OpenCV SIFT matching, with screenshot evidence.

Also built: Creator subscription API with an OpenCV blur paywall and S3 delivery (FastAPI) · AI NDIS report generator (OpenAI, wkhtmltopdf) · fee-aware crypto arbitrage scanner across 6 exchanges (ccxt) · IG Markets streaming signal service (Lightstreamer, Flask-SocketIO) · economic-calendar ETL and API (cloudscraper, PostgreSQL) · EVM wallet onboarding API (BIP-39, eth-account) · PGA TOUR GraphQL data API.

EDUCATION & CERTIFICATION

Intermediate · Karachi Board

AIC: Artificial Intelligence · Presidential Initiative for Artificial Intelligence and Computing (PIAIC)